

7.2 Orthogonal Diagonalization

In this section we will be concerned with the problem of diagonalizing a symmetric matrix A . As we will see, this problem is closely related to finding an orthonormal basis for $\mathbb{R}^n$ that consists of eigenvectors of A . Problems of this type are important because many of the matrices that arise in applications are symmetric.

The Orthogonal Diagonalization Problem

Recall from Section 5.2 that two square matrices A and B are *similar* if there is an invertible matrix P such that $P^{-1}AP = B$. Here we focus on the special case where P is orthogonal.

Definition 1

Square matrices A and B are **orthogonally similar** if there is an orthogonal matrix P such that $B = P^TAP$. If A is orthogonally similar to a diagonal matrix D , we say A is **orthogonally diagonalizable** and that P **orthogonally diagonalizes** A .

If $P^TAP = D$, multiplying left by P and right by P^T and using $PP^T = I$ yields

$$A = PDP^T \tag{2}$$

Transposing both sides and noting a diagonal matrix satisfies $D^T = D$:

$$A^T = (PDP^T)^T = (P^T)^T D^T P^T = PDP^T = A$$

so A must be symmetric. This proves one direction of the following theorem.

Conditions for Orthogonal Diagonalizability

Theorem 7.2.1

If A is an $n \times n$ matrix with real entries, the following are equivalent:

- (a) A is orthogonally diagonalizable.
- (b) A has an orthonormal set of n eigenvectors.
- (c) A is symmetric.

Proof.

(a) $\Rightarrow$ (b) If A is orthogonally diagonalizable, there is an orthogonal P with $P^{-1}AP$ diagonal. As shown in Formula (2) in the proof of Theorem 5.2.1, the n column vectors of P are eigenvectors of A . Since P is orthogonal, those columns are orthonormal, so A has n orthonormal eigenvectors.

(b) $\Rightarrow$ (a) Suppose A has an orthonormal set of n eigenvectors $\{\mathbf{p}_1, \dots, \mathbf{p}_n\}$. The matrix P with these as columns diagonalizes A (Theorem 5.2.1). Since the eigenvectors are orthonormal, P is orthogonal, so P orthogonally diagonalizes A .

(a) $\Rightarrow$ (c) Shown above: $P^T A P = D$ implies $A = P D P^T$, which gives $A^T = P D^T P^T = P D P^T = A$.

(c) $\Rightarrow$ (a) The proof that every real symmetric matrix is orthogonally diagonalizable is beyond the scope of this text; the structure of the argument is outlined in Exercise 31. ■

Properties of Symmetric Matrices

To devise an algorithm for orthogonally diagonalizing a symmetric matrix, we need the following critical properties.

Theorem 7.2.2

If A is a symmetric matrix with real entries, then:

- (a) The eigenvalues of A are all real numbers.
- (b) Eigenvectors from different eigenspaces are orthogonal.

Part (a) requires results about complex vector spaces and is discussed in Section 7.5.

Proof of (b). Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors of A for distinct eigenvalues λ_1 and λ_2 . Using Formula (26) of Section 3.2 and the symmetry $A = A^T$:

$$A\mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_1 \cdot A^T \mathbf{v}_2 = \mathbf{v}_1 \cdot A\mathbf{v}_2 \quad (3)$$

Since $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$, equation (3) gives $\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_1 \cdot \lambda_2 \mathbf{v}_2$, i.e.,

$$(\lambda_1 - \lambda_2)(\mathbf{v}_1 \cdot \mathbf{v}_2) = 0 \quad (4)$$

Since $\lambda_1 \neq \lambda_2$, we conclude $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. ■

Theorem 7.2.2 yields the following procedure. Its correctness follows because eigenvectors from *different* eigenspaces are already orthogonal (by Theorem 7.2.2(b)), and Gram–Schmidt makes eigenvectors within the *same* eigenspace orthonormal.

Orthogonally Diagonalizing an $n \times n$ Symmetric Matrix

Step 1. Find a basis for each eigenspace of A .

Step 2. Apply the Gram–Schmidt process to each of those bases to obtain an orthonormal basis for each eigenspace.

Step 3. Form the matrix P whose columns are the vectors constructed in Step 2. This matrix orthogonally diagonalizes A , and the diagonal of $D = P^T A P$ lists the eigenvalues in the same order as their corresponding eigenvectors in P .

EXAMPLE 1 | Orthogonally Diagonalizing a Symmetric Matrix

Find an orthogonal matrix P that diagonalizes

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

SOLUTION: The characteristic equation is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 4 & -2 & -2 \\ -2 & \lambda - 4 & -2 \\ -2 & -2 & \lambda - 4 \end{bmatrix} = (\lambda - 2)^2(\lambda - 8) = 0$$

so the distinct eigenvalues are $\lambda = 2$ (mult. 2) and $\lambda = 8$ (mult. 1).

Eigenspace for $\lambda = 2$: Solving $(2I - A)\mathbf{x} = \mathbf{0}$ gives basis

$$\mathbf{u}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad (5)$$

Applying Gram-Schmidt to $\{\mathbf{u}_1, \mathbf{u}_2\}$ yields orthonormal eigenvectors (verify):

$$\mathbf{v}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix} \quad (6)$$

Eigenspace for $\lambda = 8$: Solving $(8I - A)\mathbf{x} = \mathbf{0}$ gives basis $\mathbf{u}_3 = [1, 1, 1]^T$. Normalizing:

$$\mathbf{v}_3 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

Using $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ as column vectors:

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

which orthogonally diagonalizes A . One can verify (verify):

$$P^T A P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Spectral Decomposition

If A is a symmetric matrix orthogonally diagonalized by $P = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_n]$ with eigenvalues $\lambda_1, \dots, \lambda_n$, then $D = P^T A P$ and $A = P D P^T$. Multiplying out:

$$A = P D P^T = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} = [\lambda_1 \mathbf{u}_1 \ \cdots \ \lambda_n \mathbf{u}_n] \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}$$

which gives the **spectral decomposition** of A :

$$\boxed{A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T} \quad (7)$$

Each term $\lambda_k \mathbf{u}_k \mathbf{u}_k^T$ is λ_k times the standard matrix for the orthogonal projection onto the eigenspace spanned by $\mathbf{u}_k$. Thus the image $A\mathbf{x}$ equals the sum of the scaled orthogonal projections of $\mathbf{x}$ onto the eigendirections.

EXAMPLE 2 | A Geometric Interpretation of a Spectral Decomposition

The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$ has eigenvalues $\lambda_1 = -3$ and $\lambda_2 = 2$ with eigenvectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad (\text{verify}).$$

Normalizing: $\mathbf{u}_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$.

The spectral decomposition is:

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} &= (-3) \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix} + (2) \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \\ &= (-3) \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix} + (2) \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix} \end{aligned} \quad (8)$$

For $\mathbf{x} = (1, 1)^T$, formula (8) gives two ways to compute $A\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad (9)$$

and via the spectral decomposition (projecting $\mathbf{x}$ onto the eigenspaces, scaling, and summing):

$$\begin{aligned} A\mathbf{x} &= (-3) \underbrace{\begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{= \left(-\frac{1}{5}, \frac{2}{5}\right)^T} + (2) \underbrace{\begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{= \left(\frac{6}{5}, \frac{3}{5}\right)^T} = \begin{bmatrix} \frac{3}{5} \\ -\frac{6}{5} \end{bmatrix} + \begin{bmatrix} \frac{12}{5} \\ \frac{6}{5} \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \end{aligned} \quad (10)$$

The geometric picture is shown in Figure 7.2.1: the image $A\mathbf{x} = (3, 0)$ is obtained by projecting $\mathbf{x} = (1, 1)$ onto each eigenspace, scaling by the corresponding eigenvalue, and adding the results.

The Nondiagonalizable Case

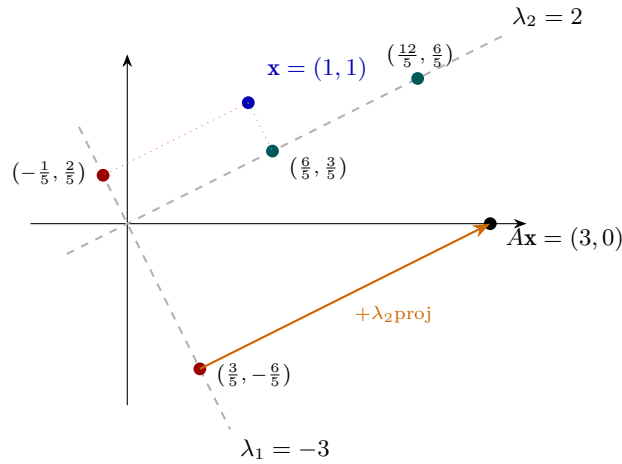


Figure 1: Spectral decomposition: $A\mathbf{x}$ as the sum of scaled eigenspace projections (red: $\lambda_1 = -3$ contribution; teal: $\lambda_2 = 2$ contribution).

If A is an $n \times n$ matrix that is not orthogonally diagonalizable, it may still be possible to achieve considerable simplification in $P^T A P$ by choosing P appropriately. The following two theorems, stated without proof, illustrate this.

Theorem 7.2.3 — Schur's Theorem

If A is an $n \times n$ matrix with real entries and real eigenvalues, then there is an orthogonal matrix P such that $P^T A P$ is an upper triangular matrix of the form

$$P^T A P = \begin{bmatrix} \lambda_1 & \times & \times & \cdots & \times \\ 0 & \lambda_2 & \times & \cdots & \times \\ 0 & 0 & \lambda_3 & \cdots & \times \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} \quad (11)$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A repeated according to multiplicity. The factorization $A = P S P^T$ (where S denotes the upper triangular matrix) is called a **Schur decomposition** of A .

Theorem 7.2.4 — Hessenberg's Theorem

If A is an $n \times n$ matrix with real entries, then there is an orthogonal matrix P such that $P^T A P$ is a matrix of the form

$$P^T A P = \begin{bmatrix} \times & \times & \cdots & \times & \times & \times \\ \times & \times & \cdots & \times & \times & \times \\ 0 & \times & \ddots & \times & \times & \times \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \times & \times & \times \\ 0 & 0 & \cdots & 0 & \times & \times \end{bmatrix} \quad (13)$$

in which every entry below the *first subdiagonal* is zero. Such a matrix is said to be in **upper Hessenberg form**. The factorization $A = P H P^T$ (where H denotes the upper Hessenberg matrix) is called an **upper Hessenberg decomposition** of A .

Remark. Unlike the diagonal entries in the Schur form (11), the diagonal entries in the Hessenberg form (13) are usually *not* the eigenvalues of A . In many numerical algorithms the matrix is first converted to upper Hessenberg form to reduce computation in subsequent steps.

Exercise Set 7.2

In Exercises 1–6, find the characteristic equation of the given symmetric matrix, then determine by inspection the dimensions of the eigenspaces.

1. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$

In Exercises 7–14, find a matrix P that orthogonally diagonalizes A , and determine $P^{-1} A P$.

7. $A = \begin{bmatrix} 6 & 2\sqrt{3} \\ 2\sqrt{3} & 7 \end{bmatrix}$

9. $A = \begin{bmatrix} -2 & 0 & -36 \\ 0 & -3 & 0 \\ -36 & 0 & -23 \end{bmatrix}$

11. $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

In Exercises 15–18, find the spectral decomposition of the matrix.

15. $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

17. $\begin{bmatrix} -3 & 1 & 2 \\ 1 & -3 & 2 \\ 2 & 2 & 0 \end{bmatrix}$

In Exercises 19–20, determine whether there exists a 3×3 symmetric matrix whose eigenvalues are $\lambda_1 = -1$, $\lambda_2 = 3$, $\lambda_3 = 7$ and for which the corresponding eigenvectors are as stated. Find the matrix if it exists; if not, explain why.

19. $\mathbf{x}_1 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

21. Let A be a diagonalizable matrix with the property that eigenvectors corresponding to distinct eigenvalues are orthogonal. Must A be symmetric? Explain.

22. Assuming $b \neq 0$, find a matrix A that orthogonally diagonalizes $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$.

In Exercise 23, let $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be multiplication by A . Find two orthogonal unit vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ such that $T_A(\mathbf{u}_1)$ and $T_A(\mathbf{u}_2)$ are orthogonal.

23. a. $A = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$ b. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Working with Proofs

25. Prove that if A is any $m \times n$ matrix, then $A^T A$ has an orthonormal set of n eigenvectors.

28. a. If $\mathbf{v}$ is any $n \times 1$ matrix and I is the $n \times n$ identity matrix, prove that $I - \mathbf{v}\mathbf{v}^T$ is orthogonally diagonalizable.

b. Find a matrix P that orthogonally diagonalizes $I - \mathbf{v}\mathbf{v}^T$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

29. Prove that if A is a symmetric orthogonal matrix, then 1 and -1 are the only possible eigenvalues.